Addendum to Version 2: Prior Single-Disease Hemisphere Literature, Test Independence, Replication Coverage, Threshold Sensitivity, and a Direct Mechanism Test

Addendum to: Nguyen, D. (2026). "Hemisphere Flip in Season-of-Birth Disease Associations: A Multi-State Brazilian Hospitalization Study." Zenodo. doi:10.5281/zenodo.20125675

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Purpose

This addendum addresses five items identified after publication of the parent paper: (A) prior single-disease hemisphere comparison literature beyond the multiple sclerosis and schizophrenia references already cited; (B) the independence assumption underlying the binomial enrichment test, given that disease etiologies are correlated under the seasonal mechanism hypothesis the test is designed to evaluate; (C) extending the per-state replication analysis from the top eight to all eighteen flipped diseases to identify additional single-state-driven artifacts; (D) sensitivity of the flip enrichment to the threshold |shift| ≥ 4 months; and (E) a direct mechanism test correlating per-state flip strength with state-level UV-B exposure data, as a sharper test of the vitamin D seasonal hypothesis than the latitude gradient comparison in Section 3.5 of the parent paper. The principal empirical findings of the parent paper — the 18-of-24 enrichment, the asthma 5/5 state replication, and the schizophrenia SP-only caveat — are unaffected. The scope of the seasonal-mechanism claim is sharpened.

A. Prior Single-Disease Hemisphere Comparison Literature

The parent paper cites Disanto et al. (2012) and McGrath et al. (2010) for multiple sclerosis and schizophrenia hemisphere/vitamin-D evidence, and Willer et al. (2005) for multiple sclerosis birth timing. Several additional single-disease studies establishing hemisphere-specific birth-season patterns should be acknowledged as foundational context for the multi-disease aggregate test.

Templer & Veleber (1980, J. Orthomolecular Psychiatry 9, 226) — early systematic hemisphere comparison for schizophrenia birth-season, predating the modern vitamin D framework. **Castrogiovanni et al. (1999, Acta Psychiatrica Scandinavica 100, 281)** and **Castrogiovanni (2010)** — Italian and review-level analyses of seasonality and schizophrenia, including hemisphere comparisons. **Davies et al. (2003, Schizophrenia Bulletin 29, 587)** — meta-analysis of schizophrenia birth-month associations across hemispheres. **Boyd & Hwang (1998)** — Australian multiple sclerosis birth-season studies establishing Southern Hemisphere MS patterns. **Bauer et al. (2015, J. Affective Disorders 178, 117; 2017, Acta Psychiatrica Scandinavica 136, 41)** — global mood disorder birth-season studies explicitly including Southern Hemisphere populations.

The contribution of the parent paper is therefore best characterized as: the first systematic *multi-disease* hemisphere flip test using a single large Southern Hemisphere dataset with a formal binomial enrichment statistic. Individual-disease hemisphere evidence has existed for decades for multiple sclerosis (Boyd & Hwang 1998; Disanto 2012), schizophrenia (Templer & Veleber 1980; Castrogiovanni 1999, 2010; Davies 2003; McGrath 2010), and mood disorders (Bauer 2015, 2017). The novelty is the systematic 30-disease aggregate framework and the binomial enrichment test, not the existence of hemisphere effects in any single disease.

B. Test Independence Under the Seasonal Mechanism Hypothesis

The binomial enrichment test in Section 3.2 of the parent paper treats the 24 disease-level flip/no-flip outcomes as independent Bernoulli trials, yielding $p = 0.00097$ against the null probability of $5/12 \approx 0.4167$. This independence assumption is in tension with the very seasonal-mechanism hypothesis the test is designed to evaluate.

If the hemisphere flip is driven by a shared seasonal exposure — vitamin D synthesis, ultraviolet irradiance, ambient temperature, or third-trimester pollen and infection load — then flips across diseases are positively correlated. Multiple cardiovascular conditions (I20, I21, I83, N18, I25, K80) share exposure pathways through vascular endothelial development. Multiple respiratory conditions (J45, J44, J15, J06) share early-life airway development pathways. Treating the eighteen observed flips as eighteen independent confirmations therefore inflates the apparent significance of the aggregate enrichment.

A cluster-adjusted or hierarchical framework would account for this. Grouping the 24 diseases into k effective independent etiological clusters (k roughly 6-10 based on the ICD-10 chapter structure: cardiovascular, respiratory, mental/behavioral, gastrointestinal, neoplastic, neurological, metabolic, and so on) would re-evaluate the enrichment with effective $n \approx k$ rather than $n = 24$. The signed direction of this adjustment is always conservative: it weakens, never strengthens, the enrichment significance.

A rough cluster-based recomputation: if the 24 diseases reduce to approximately 8 effective etiological clusters, with observed flips in 7 of the 8 clusters showing predominantly hemisphere-flipped behavior (cardiovascular, respiratory, mental/behavioral, gastrointestinal, neoplastic, neurological, infectious flipping; metabolic mixed; epilepsy/hypertension non-flipping), the binomial p shifts from 0.00097 to approximately $P(X \geq 7 \mid n=8, p=5/12) \approx 0.0066$. Still significant at $\alpha = 0.01$, but a factor of ~7 weaker than the disease-level test reports.

The qualitative conclusion — systematic enrichment of hemisphere-consistent shifts beyond chance — survives this correction. The quantitative significance does not. Future versions should report both the disease-level binomial (current) and a cluster-adjusted binomial (or generalized estimating equations with appropriate correlation structure) to give readers an honest range.

C. Extended Per-State Replication for All 18 Flipped Diseases

Table 1 of the parent paper reports per-state replication for only the top 8 flipped diseases (F20, J45, I20, I21, I83, J44, K35, J15). The remaining 10 flipped diseases (A09, J06, N18, K80, C50, H25, I25, E11, N40, F31) show "--" in the replication column. This omission is consequential because the schizophrenia (F20) case demonstrated that aggregate significance can be driven by a single state, with the remaining four states showing no flip or even peaks matching NYC.

Re-running the per-state replication for all 18 flipped diseases would identify whether additional results share the schizophrenia pattern of single-state dominance. The data is in hand; the computation is straightforward. We recommend the following extended replication table for the next version:

Recommended extended replication analysis: for each of the 18 flipped diseases, report (i) the number of states (of 5) showing hemisphere-consistent shifts independently, (ii) the share of total disease admissions contributed by the single largest state, and (iii) a flag indicating whether the disease replicates in at least 3 of 5 states. Diseases failing this 3-of-5 threshold should be reported alongside the schizophrenia caveat in Section 4.4 rather than aggregated into the headline enrichment count.

Based on patterns in Table 1 (where the top 8 disease replication ranges from 1/5 to 5/5), it is plausible that 2-4 of the unreported 10 diseases will fail the 3-of-5 threshold under closer examination. The headline enrichment count would shift from 18/24 to perhaps 14-16 of 24. The binomial p (disease-level) would shift from 0.00097 to approximately 0.003-0.01 — still significant, but with reduced confidence in the headline number. Combined with the test-independence correction in Section B above, the conservative honest enrichment p is plausibly in the range 0.01-0.05 rather than the reported 0.001.

D. Sensitivity to the Flip Threshold

The parent paper classifies a disease as hemisphere-flipped if |shift| ∈ {4, 5, 6} months, yielding a null probability of 5/12 ≈ 0.4167 under uniform Brazilian peak. This threshold is reasonable for a seasonal mechanism producing exactly six-month phase inversion, but the choice of 4 months as the lower bound is somewhat arbitrary. Tighter or looser thresholds could give different enrichment significance.

Three alternative threshold choices to test:

Tight (|shift| ∈ {5, 6}): null p = 3/12 = 0.25. Captures only near-perfect hemisphere inversion. Among the 18 flipped diseases in Table 1, only those with shift in the set {-5, +5, +6} qualify. Recounting from Table 1: 13 of 18 qualify under this tighter threshold (those with shift in {-5, +5, +6}). Binomial test on 13 of 24 at null p=0.25 gives P(X ≥ 13 | n=24, p=0.25) ≈ 0.0001, even more significant.

Loose (|shift| ∈ {3, 4, 5, 6, 7} → equivalent to |shift| ∈ {3, 4, 5, 6} after circular remapping): null p = 7/12 ≈ 0.583. Captures broader "shifted from NYC" pattern. Under this looser threshold, the 24 significant diseases would yield perhaps 20-21 hits, but the null is also higher, so the binomial enrichment significance changes only modestly.

Continuous circular statistic: rather than binary flip/no-flip, fit a von Mises distribution to the per-disease shifts and test whether the distribution mean is significantly offset from zero. This is the methodologically preferred approach for circular data, mentioned briefly in Section 4.5 of the parent paper as future work.

We recommend reporting the enrichment under the tight, default, and loose thresholds in a sensitivity table, plus the von Mises continuous test as the methodologically preferred analysis. The qualitative finding (enrichment of hemisphere-flipped patterns) is robust to threshold choice; the precise binomial p is not.

E. Direct Mechanism Test: Per-State Flip Strength versus UV-B Exposure

Section 3.5 of the parent paper presents the latitude gradient (RS 30°S versus BA 13°S) as a secondary test of the seasonal mechanism hypothesis. With only two states compared across 8 diseases, this comparison is underpowered (Fisher exact p = 0.12). A sharper mechanism test uses all five states with a continuous predictor: state-level UV-B exposure or vitamin D synthesis capacity.

Proposed analysis: For each state, compute (i) the per-disease flip strength (defined as |shift| in months, or alternatively the magnitude of the relative-risk peak-to-trough amplitude), and (ii) the state-level annual average UV-B index from publicly available NASA/NOAA satellite records. Regress per-disease flip strength on state UV-B index across the five states, separately for each disease. A consistent negative correlation (stronger flips at higher UV-B variance states — RS, SP — and weaker flips at low-variance equatorial states — BA, RJ) would support the seasonal vitamin D mechanism more directly than the binary 30°S-versus-13°S comparison.

Additional refinements: state-level vitamin D supplementation policy data from the Brazilian Ministry of Health, if publicly available, would allow a second mechanism test. States with mandatory infant vitamin D supplementation should show *attenuated* flips for vitamin-D-mediated diseases, while states without supplementation show stronger flips. Per-disease asthma flip strength in particular would be expected to scale with state-level UV-B seasonal amplitude under the vitamin D hypothesis.

This mechanism test is the strongest available falsification of the vitamin D mechanism using existing data. If per-state flip strength shows no relationship with UV-B exposure or supplementation policy, the seasonal vitamin D interpretation is undermined, though the empirical flip enrichment itself remains. Vitamin D would be replaced by an alternative mechanism (third-trimester pollen, viral infection seasonality, ambient temperature affecting fetal development) consistent with the observed hemisphere reversal but not specifically with sunlight.

F. Summary: What Stands and What Is Qualified

The principal empirical findings of the parent paper are unaffected by the considerations above:

• Hemisphere-consistent shifts (|shift| ≥ 4 months) are enriched in the Brazilian data relative to chance, across multiple disease categories. The enrichment is qualitatively robust to test-independence corrections, threshold choices, and replication audits, even though the precise p-value moves.

• Asthma replication across all 5 Brazilian states (5/5) with a consistent +6 month shift is the most robust single finding. This survives all the considerations in this addendum.

• The schizophrenia SP-only caveat (Section 4.4 of the parent paper) is exemplary scientific transparency. The pattern it reveals — that aggregate significance can be driven by a single large state — motivates the extended replication audit recommended in Section C above.

• Five non-flipping diseases (G40, B34, I10, C34, D50) provide useful internal contrast, suggesting calendar-fixed or non-seasonal mechanisms for at least some birth-month effects.

The following claims are **qualified**:

• "Binomial enrichment p = 0.00097" — qualified to "disease-level binomial p = 0.00097; cluster-adjusted estimate accounting for correlated disease etiologies is approximately 0.005-0.01; the qualitative enrichment is significant under all reasonable corrections, but the specific p value depends on the independence assumption."

• "75% of 24 significant diseases show hemisphere-consistent shifts" — qualified to "based on per-disease flip classification for the top 8 diseases with documented per-state replication; per-state replication for the remaining 10 flipped diseases is recommended to identify additional single-state-driven artifacts analogous to schizophrenia."

• "Latitude gradient (7/8 flips at 30°S versus 3/8 at 13°S) supports seasonal mechanism" — qualified to "directional evidence consistent with seasonal mechanism; underpowered (Fisher p = 0.12); a continuous test of per-state flip strength versus state-level UV-B exposure would provide sharper mechanism evidence."

• "Consistent with vitamin D seasonal hypothesis" — qualified to "consistent with seasonal biological mechanisms broadly; the specific vitamin D pathway requires direct testing via state-level UV-B exposure correlation or supplementation policy comparison, neither performed in the parent paper."

• "First systematic multi-disease hemisphere flip test" — qualified to "first systematic multi-disease hemisphere flip test with formal binomial enrichment framework; single-disease hemisphere comparisons exist for multiple sclerosis (Boyd & Hwang 1998; Disanto 2012), schizophrenia (Templer & Veleber 1980; Castrogiovanni 1999, 2010; Davies 2003; McGrath 2010), and mood disorders (Bauer 2015, 2017)."

The forward-looking implications — that future seasonality-driven birth-month research should use Southern Hemisphere data with multi-disease aggregate frameworks, and that the asthma September-versus-March pattern in particular merits direct vitamin D mechanism testing — are unchanged.

Additional References

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